



**Verified Carbon
Standard**

GUANGDONG LANKOU 26MW HYDRO POWER PROJECT



Document Prepared By LGAI Technological Center, S.A. (Applus+
Certification)

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Work Carried Out By	Lead Auditor / Technical Expert: Doris Dai Technical Reviewer: Simon Shen

Summary:

LGA Technological Center, S.A. (hereafter referred to as “Applus+ Certification”) has been commissioned by Beijing 3050 Technology Co.,Ltd. to perform the verification of greenhouse gas emission reductions of the project activity “Guangdong Lankou 26MW Hydro Power Project” (VCS Ref. No. 371, hereafter referred to as “the project activity”) reported in the monitoring report /1/ during monitoring period 01/06/2012 to 31/10/2017.

The project activity has been validated by JCI based on the CDM PDD /3/ version 5.0 dated 01/03/2009 and reported in the validation report No. JCI-VAL 07/036 /4/, version 05, completed on 07/07/2009. The project activity was registered as a CDM project activity on 07/07/2009 which is available at <https://cdm.unfccc.int/Projects/DB/JCI1238492309.45/view>. A gap validation was performed by CEC during the VCS verification which is available at <https://registry.terra.org/app/projectDetail/VCS/371>. The CDM PDD /3/ would be the basis for verification.

The project activity is a hydropower project located at Bainitang Village, Lankou Town, Dongyuan County, Heyuan City, Guangdong Province, P. R. China which is to use hydropower resource for electricity generation. The installed capacity of the project activity is 26 MW, involves the installation of 3 sets of hydro turbine generators with a unit capacity of 8.66 MW. The average annual power delivered to the grid by the project is expected to be 80,070 MWh. The project can reduce GHG emissions by replacing the electricity generated by fossil fuel fired power plants in China Southern Power Grid (CSPG). It's estimated that the proposed project could achieve GHG emission reductions of 69,757 tCO₂e annually.

The purpose and scope of this verification is to ensure that reported emission reductions are complete and accurate in accordance with applicable VCS standards and relevant UNFCCC requirements in order

to be certified. A desk review and a site visit have been conducted to verify the data submitted in the monitoring report /1/. Applus+ Certification confirms the following has been reviewed:

- Monitoring plan included in the registered CDM PDD /3/ version 5.0 dated 01/03/2009;
- Validation report No. JCI-VAL 07/036 /4/ version 05, dated 07/07/2009;
- VCS Verification report No. 09016020 /4/, dated 07/01/2010;
- Approved methodology, ACM0002 /7/, version 7.0, dated 14/12/2007;
- VCS standards version 4.3 and guidance version 4.2, as well as relevant UNFCCC requirements;
- All information and references relevant to the project activity's resulting in emission reductions.

During this verification, no finding was identified related to the monitoring, implementation or operations of the project activity in relation to relevant VCS standards, guidance and UNFCCC requirements and relevant host party criteria and the applied baseline and monitoring methodology etc.

Applus+ Certification confirms that the project is implemented in accordance with the registered PD /3/. The monitoring plan complies with the applied methodology ACM0002 /7/ version 7.0 and the monitoring has been carried out in accordance with the registered PD. The monitoring system is in place and the emission reductions are calculated without material misstatements. The level of assurance of the verification is reasonable. Our opinion relates to the projects GHG emissions and the resulting GHG emission reductions reported and related to the valid and registered project baseline and monitoring and its associated documents. Based on the information reviewed and evaluated Applus+ Certification confirms that the implementation of the project has resulted in 366,109 tCO₂e emission reductions during period 01/06/2012 to 31/10/2017.

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1 INTRODUCTION

1.1 Objective

- LGAI Technological Center, S.A. (Applus+ Certification) has been commissioned by Beijing 3050 Technology Co.,Ltd. to perform the verification of greenhouse gas emission reductions of the project activity “Guangdong Lankou 26MW Hydro Power Project” (VCS Ref. No. 371) reported in the Monitoring Report /1/ during monitoring period 01/06/2012 to 31/10/2017.
- LGAI Technological Center, S.A. (Applus+ Certification) as the verification body of the project activity has been accredited as a DOE by UNFCCC and also meets the competence requirements as set out in ISO 14065:2020.
- The objective of verification is to have an independent review and ex post determination by a Validation and Verification Body (VVB) of the monitored reductions in GHG emissions that have occurred as a result of the registration of VCS project. Certification is the written assurance by the VVB that, during a specific time period, a proposed VCS project activity achieved the reductions in anthropogenic emissions by sources of GHGs as verified.
- The objective of this verification/certification is to verify and certify emission reductions, reported for the “Guangdong Lankou 26MW Hydro Power Project” in China for the period 01/06/2012 to 31/10/2017.

1.2 Scope and Criteria

- The verification scope is defined as an independent and objective review of the registered PD and PDD, the Project’s baseline study and Monitoring Report (MR) and other relevant documents. The information in these documents is reviewed against VCS Version 4.3 requirements, UNFCCC rules and associated interpretations.
- The verification is not meant to provide any consulting towards the client. However, stated requests for forward actions and/or corrective actions may provide input for improvement of the Project monitoring towards reductions in the GHG emissions.

1.3 Level of Assurance

- The verification report is based on the CDM-PDD, the VCS Monitoring Report (MR), supporting evidences made available to the verifier and information collected through performing interviews and during the on-site assessment.
- The verification conclusion is assured a reasonable level of assurance.

1.4 Summary Description of the Project

Project title	Guangdong Lankou 26MW Hydro Power Project
VCS reference number	371
Project Participants	Shenzhen City Fuyuan Industry (Group) Co., Ltd. (Project Owner, host country, P. R. China)
Location of the project	Bainitang Village, Lankou Town, Dongyuan County, Heyuan City, Guangdong Province, People's Republic of China Geographic coordinates: East longitude: 115°02'44" North latitude: 23°55'56"
Project start date	Construction start date: 04/12/2005 Operation start date: 01/11/2007
Version of CDM PDD	Version 5.0, dated 01/03/2009 (basis for verification)
Monitoring period	01/06/2012 to 31/10/2017
First monitoring report	Version 01, dated 20/05/2022
Final monitoring report	Version 02, dated 14/07/2022
Applied Methodology/Version	ACM0002, version 7.0, dated 14/12/2007
Scope/Technical Area	1/1.2

The project activity is a hydropower project located at Bainitang Village, Lankou Town, Dongyuan County, Heyuan City, Guangdong Province, P. R. China which is to use hydropower resource for electricity generation. The installed capacity of the project activity is 26 MW, involves the installation of 3 sets of hydro turbine generators with a unit capacity of 8.66 MW. The average annual power delivered to the grid by the project is expected to be 80,070 MWh. The project can reduce GHG emissions by replacing the electricity generated by fossil fuel fired power plants in China Southern Power Grid (CSPG). It's estimated that the proposed project could achieve GHG emission reductions of 69,757 tCO₂e annually.

The project activity has been validated by JCI based on the CDM PDD /3/ version 5.0 dated 01/03/2009 and reported in the validation report No. JCI-VAL 07/036 /4/, version 05, completed on 07/07/2009. The project activity was registered as a CDM project activity on 07/07/2009 which is available at <https://cdm.unfccc.int/Projects/DB/JCI1238492309.45/view>. A gap validation was performed by CEC during the VCS verification which is available at <https://registry.terra.org/app/projectDetail/VCS/371>. The CDM PDD /3/ would be the basis for verification.

2 VERIFICATION PROCESS

2.1 Method and Criteria

Verification was conducted using Applus+ Certification's procedures in line with the requirements specified in the VCS Standard version 4.3, CDM M&P, the latest version of the CDM Validation and Verification Standard, and relevant UNFCCC requirements and applying standard auditing techniques.

Applus+ Certification completed a strategic review and risk assessment of the projects activities and processes in order to gain a full understanding of:

- Activities associated with all the sources contributing to the project emissions and emission reductions, including leakage if relevant;
- Protocols used to estimate or measure GHG emissions from these sources;
- Collection and handling of data;
- Controls on the collection and handling of data;
- Means of verifying reported data; and
- Compilation of the monitoring report.

Applus+ Certification verified the implementation of the monitoring plan and the data presented in the Monitoring Report /1/ for the period in question. This involved a site visit and a desk review of the Monitoring Report. This Verification Report describes the findings of this assessment.

The information of the assessment team is included in below:

Assessment team

According to the sectoral scopes / technical area and experiences in the sectoral or national business environment, Applus+ Certification has composed a project assessment team in accordance with the appointment rules in Applus+ Certification. The composition of assessment team has to be approved by the Applus+ Certification ensuring that the required skills are covered by the team. The four qualification levels for team members that are assigned by formal appointment rules as below:

- Leader Auditor (LA)
- Auditor (A)/ Auditor Trainee (AiT)

- Technical Reviewer (TR)
- Technical Experts (TE)

Name	Qualification	Coverage of scope	Coverage of Technical Area	Host country experience
Doris Dai	LA/TE	Y (1.2)	Y	Y
Simon Shen	TR	Y (1.2)	Y	Y

- **Doris Dai** (Master's Degree in Environmental Sciences, Bachelor's Degree in Environmental Technology) is an Auditor appointed by Applus+ LGAI for the GHG project assessment and auditing. She has more than 6 years of work experience in CDM/VCS project assessment. Before she joined Applus+ LGAI, she has been working for CTI Certification as senior GHG Auditor for 3.5 years.
- **Simon Shen** (Master Degree in Thermal Energy Engineering, Bachelor Degree in Environmental Engineering) is a Lead Auditor appointed by Applus+ Certification for the GHG project assessment. He is based in Shanghai. He has several years of work experience in environmental protection field. Before he joined Applus+ Certification, he had been worked for TÜV SÜD as a GHG Validator/Assessment team and ISO 9001/14001 Lead Auditor for 3.5 years.

2.2 Document Review

- The VCS monitoring report /1/ version 01 dated 20/05/2022, version 02 dated 14/07/2022 and the emission reduction calculations spreadsheet /2/, were assessed as part of the verification. In addition, the CDM PDD /3/ version 5.0 dated 01/03/2009 in particular the baseline estimations and the monitoring plan, the CDM Validation Report /4/ dated 07/07/2009, as well as relevant documents, were reviewed. A detailed document reviewed are listed in Appendix 1 of the report.

2.3 Interviews

The key personnel interviewed are summarized in the table below:

Interviewed personnel	Role	Organization	Subject
Mr. Qiu Guocheng	Director	Shenzhen City Fuyuan Industry (Group) Co., Ltd.	Operation of the project activity; Implementation of the monitor plan of the project activity; Data collection and data achievement; Calibration of meters and equipment maintenance.
Mr. Chen Hanliang	Operator	Shenzhen City Fuyuan Industry (Group) Co., Ltd.	
Mr. Su Xianran	Operator	Shenzhen City Fuyuan Industry (Group) Co., Ltd.	
Mr. Xie Tingheng	Villager	Xiewu Village	
Mr. Chen Yuzhou	Villager	Shiba Village	
Ms. Gu Fuhua	Villager	Qikeng Village	
Mr. Chen Ting	Local Environmental Bureau	Officer	
Mr. Zheng Jiangzhuo	Beijing 3050 Technology Co.,Ltd.	Project Manager	Data collection and ER calculation.

2.4 Site Inspections

- The assessment team performed the on-site verification (Bainitang Village, Lankou Town, Dongyuan County, Heyuan City, Guangdong Province, P. R. China) on 13/07/2022. The interviewed personnel and objective are listed in above table.

2.5 Resolution of Findings

- As an outcome of the verification process, the team can raise different types of findings.
- Where a non-conformance arises the assessment team shall raise a Corrective Action Request (CAR). A CAR is issued, where:
 - a) Non-compliance with the monitoring plan or methodology are found in monitoring and reporting and has not been sufficiently documented by the project participants, or if the evidence provided to prove conformity is insufficient;
 - b) Modifications to the implementation, operation and monitoring of the project activity has not been sufficiently documented by the project participants;
 - c) Mistakes have been made in applying assumptions, data or calculations of emission reductions that will impact the quantity of emission reductions;
 - d) Issues identified in a FAR during validation to be verified during verification or previous verification(s) have not been resolved by the project participants.
- The assessment team shall raise a Clarification Request (CL) if information is insufficient or not clear enough to determine whether the applicable CDM requirements have been met.
- All CARs and CLs raised during verification shall be resolved prior to submitting a request for issuance.
- There is no CARs and CLs raised for this monitoring period for the project.

2.5.1 Forward Action Requests

- None FAR was raised during the verification process. Also there are no remaining from former validation and verification.

2.6 Eligibility for Validation Activities

- Not applicable as LGAI Technological Center, S.A. holds the accreditation for the validation and verification for projects under scope 1.

3 VALIDATION FINDINGS

3.1 Participation under Other GHG Programs

- Through reviewing the registered PDD /3/ and validation report /4/, it was validated that the project has been registered as a CDM project with reference No. 2458 first which is available at <https://cdm.unfccc.int/Projects/DB/JCI1238492309.45/view>. Then the project has been registered as VCS project which is available at <https://registry.terra.org/app/projectDetail/VCS/371>. The project does not participate in the other emissions trading program by checking public information on Internet, interviewing with project owner and statement issued by project owner.
- During the period from 01/11/2007 to 31/10/2017 as the first VCS crediting period, the project would claim only for VCU or CERs. But VCUs and CERs will not be claimed in the same period. Also, for period 07/07/2009 to 31/05/2012, the project has applied for the issuance of CERs.
- Therefore, Applus+ Certification consider the project is eligible to participate under the VCS Program as there is no double counting for the emission reduction during any period.

3.2 Methodology Deviations

- Not applicable as no deviation for methodology.

3.3 Project Description Deviations

- There were 2 deviations for the project activity during this monitoring period:
- 1. The project was registered under VCS version 2007 and completed validation before 19/03/2020. Thus, it remains eligible to apply the crediting period requirements under VCS standard Version 3 which shall be a maximum of ten years and maybe renewed at most twice, so the first renewable crediting period of the project should be updated from 01/11/2007~06/07/2009 to 01/11/2007~31/10/2017.
- 2. There was a typo for information of equipment, as in the registered PDD, the rated power of the generator is indicated as 8,166.67 kW. However, the installed capacity should be 26 MW consists of 3 sets of generators with 8,666.7 kW for each. By on-site visit and checking nameplate of the equipment, it is confirmed that the actual rated power of the generator should be 8,666.7 kW for each.
- The assessment team confirm both deviations in line with the requirement of VCS and it has no impacts on the applicability, additionality or baseline of the project.

3.4 Grouped Project

- Not applicable as not a grouped project.

4 VERIFICATION FINDINGS

4.1 Project Implementation Status

- By means of on-site visit, the assessment team confirms that all physical features of the proposed CDM project activity proposed in the registered CDM PDD /3/ are in place and the PP has operated the project as per registered CDM PDD /3/. The installed capacity of the project is 26 MW, involves the installation of 3 sets of hydro turbine generators with a unit capacity of 8.66 MW. The electricity generated is transmitted to the local Power Grid via a newly built transformer station, which was then exported to the CSPG. The project activity was expected to supply 80,070 MWh of electricity to the grid. The construction of the project started on 04/12/2005, the project has been put into operation on 01/11/2007 verified by checking information on the UNFCCC website and site visit. There are no changes on the key equipment and technology since the validation of the project. No special event which would affect the monitoring of the project has been observed during the monitoring period.
- By checking information of other GHG programs such as CDM, GS, GCC, CCER etc., it is able to confirm:
- The project has not participated or been rejected under any other GHG programs since validation or previous verification;
- The project has not received or sought any other form of environmental credit, or has become eligible to do so since validation or previous verification;
- The GHG emission reductions or removals generated by the project have not become included in an emissions trading program or any other mechanism that includes GHG allowance trading.
- The project would contribute to sustainable development in as below:
- The project has supplied 420,237.338 MWh in this monitoring period that otherwise have been produced by thermal power generation projects, and contributes to Chinese renewable energy target.
- The project generated 30 decent working opportunities for local women and men, 15 for women and 15 for men and help them improve the incomes, and contributes to the decent work and economic growth.
- The emission reduction of the project is 366,109 tCO_{2e} in this monitoring period, and contributes to Chinese Climate change target.
- Moreover by checking public information, staff roaster of project /15/, payment slip /16/ and site visit, following information has been confirmed:

Row number	SDG Target	SDG Indicator	Net Impact on SDG Indicator	Current Project Contributions	Contributions Over Project Lifetime
1)	7.2	7.2.1 Renewable energy share in the total final energy consumption	Implemented activities to increase	The project utilizes hydro energy for electricity generation, which could increase the ratio of renewable energy in total energy consumption of China. From 01/06/2012 to 31/10/2017 the electricity supplied from the project is 420,237.338 MWh.	From the operation start date of the project to the end of this monitoring period, the electricity supplied from the project is around 705,250.230 MWh.

2)	8.5	8.5 Achieve full and productive employment and decent work for all women and men	Implemented activities increase to	During this monitoring period (01/06/2012 to 31/10/2017), the project achieves full and productive employment and decent work for all women and men. In 2012, there are 30 jobs created for 15 men and 15 women, the average yearly salary is 32,360 RMB/year In 2013, there are 30 jobs created for 15 men and 15 women, the average yearly salary is 34,484 RMB/year In 2014, there are 30 jobs created for 15 men and 15 women, the average yearly salary is 36,971 RMB/year In 2015, there are 30 jobs created for 15 men and 15 women, the average yearly salary is 38,623 RMB/year In 2016, there are 30 jobs created for 15 men and 15 women, the average yearly salary is 40,826 RMB/year	From the operation start date of the project to the end of this monitoring period, the project provided 30 jobs for women and men from the year of 2007 to the year of 2017, women, and men both are worked for the project, and the average yearly salary has increased from 21,304 RMB/year in 2007 to 42,726 RMB/year in 2017. The project achieves full and productive employment and decent work for all women and men.
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				In 2017, there are 30 jobs created for 15 men and 15 women, the average yearly salary is 42,726 RMB/year	
3)	13	Tonnes of greenhouse gas emissions avoided or removed	Implemented activities to increase	By utilizing water for power generation and replaced it from thermal power generation projects From 01/06/2012 to 31/10/2017 the emission reduction from the project is around 366,109 tCO ₂ e	From the operation start date of the project to the end of this monitoring period, the emission reduction from the project is around 614,414 tCO ₂ e.

- The technical parameters have been verified with the nameplates /9 / as below:

Hydropower turbines		Generators	
Turbine Type	GZ1250a-WP-550	Type	SFWG7350-80/6220
Rated head	4.00 m	Rated power	8,666.7 kW
Rated speed	68.2 r/min	Rate factor	Cosφ = 0.9
Designed flow	768 m ³ /s	Rated voltage	6.3 kV
Lifetime	30 years	Lifetime	30 years

- By comparing the actual ER claimed in this monitoring period with the estimate in the registered PD, the actual emission reductions (366,109 tCO₂e) are lower than what is stated in the registered PDD (i.e. 378,216 tCO₂e, equals to annual emission reductions, 69,757 tCO₂e multiplied by the actual operational days (1,979 days) then divided by 365 days). The assessment team consider this variation is acceptable as conservativeness.
- Therefore, the assessment team confirmed the ER in this monitoring period is not overestimated.
- The assessment team confirmed that there is no proposed or actual change to the project design during this monitoring period.
- All required equipments and procedures are available and implemented in an appropriate manner.

- All necessary monitoring instruments are installed. All required instruments including standby and operating procedures for the same have been implemented in an appropriate manner.
- The project is completely operational and the same has been confirmed on-site. Neither mistakes nor malfunction on main meters have been observed during this monitoring period.

4.2 Safeguards

4.2.1 No Net Harm

- By checking the EIA summary and conclusion provided in the registered PDD, it is confirmed that hydro power is green power and the impact caused by hydro power on the surrounding ecosystem and residents, water, and atmosphere etc. is very little, there would be no net harm caused due to the project activity. Also, the EIA of the project are approved by the government.
- Also, no potential environment or social economic matter was found during the site visit. The project is renewable energy project and thus no net harm observed in air or water quality on-site.

4.2.2 Local Stakeholder Consultation

- A survey was carried out through the information publication, distributing and collecting responses to questionnaires targeting on local residents, builders and members of the local authorities. The project owner introduced the proposed project, and then a survey was arranged through a one-page questionnaire, which was designed to be easily filled in. The opinions expressed by the stakeholders were recorded and are available on request.
- The stakeholder meeting and the survey showed that the proposed project receives strong support from the local community. They all believe the proposed project will promote local economic development and agree with the project development and construction.
- Communications with Local stakeholders was being carried out at periodic intervals. There are no negative comments received for the project.
- All such conclusion has been verified through site visit and check registered PDD.

4.3 AFOLU-Specific Safeguards

- Not applicable as non-AFOLU project.

4.4 Accuracy of GHG Emission Reduction and Removal Calculations

- The monitoring has been carried out in accordance with the monitoring plan contained in the CDM PDD /3/. All parameters were monitored and determined as per the monitoring plan which is listed in below table:

Data / Parameter:	EG _{output,y}														
Data unit:	MWh														
Description:	Annual on-grid electricity supplied to CSPG by the proposed project														
Purpose of the data:	Calculation of baseline emissions														
Parameter value:	420,608.560														
Source of data used:	Annual on-grid electricity supplied to CSPG by the proposed project (EG _{output,y}) are all sourced from Meter Reading Record (MRRs) /10/ issued by the project owner and Electricity Transaction Notes (ETNs) /11/ issued by power grid company covering monitoring period.														
Information flow:	2 bidirectional electricity meters (M1 as main and M2 as backup) installed at the entrance of the Lankou 110kV Transformer Station and the exit of the booster station of the project respectively were measured continuously, recorded monthly and archived electronically. At 24:00 hr the last day of each month, the staff from project owner will record main electricity meter's readings and form Meter Reading Records (MRRs). The staff from power grid company will record the meter readings of meters (Main meter) then transcribes the data into Electricity Transaction Notes (ETNs). The data for MRRs and ETNs have been sent to the CDM consulting company for reporting of GHG emission reduction. The conservative one would be used for ER calculation.														
Monitoring method, frequency and equipments:	The parameter was measured continuously and recorded monthly by 2 bidirectional electricity meters (M1 as main and M2 as backup) installed at the entrance of the Lankou 110kV Transformer Station and the exit of the booster station of the project respectively during the monitoring period verified by site visit. See below for the information of 2 bidirectional electricity meters verified by site visit and checking calibration certificates /12/: <table border="1"> <thead> <tr> <th>Meter</th><th>Type</th><th>Serial Number</th><th>Accuracy</th></tr> </thead> <tbody> <tr> <td>M1</td><td>DTSD341</td><td>01010801048</td><td>0.5s</td></tr> <tr> <td>M2</td><td>YD2103</td><td>070817234</td><td>0.5s</td></tr> </tbody> </table> The type, serial number and accuracy have been confirmed by site visit.			Meter	Type	Serial Number	Accuracy	M1	DTSD341	01010801048	0.5s	M2	YD2103	070817234	0.5s
Meter	Type	Serial Number	Accuracy												
M1	DTSD341	01010801048	0.5s												
M2	YD2103	070817234	0.5s												

Calibration:	The calibration information is shown as below /12/:		
	Meter	Calibration date	Valid until
	M1	12/04/2012	11/04/2013
		11/04/2013	10/04/2014
		10/04/2014	09/04/2015
		09/04/2015	08/04/2016
		08/04/2016	07/04/2017
		07/04/2017	06/04/2018
	M2	12/04/2012	11/04/2013
		11/04/2013	10/04/2014
		10/04/2014	09/04/2015
		09/04/2015	08/04/2016
		08/04/2016	07/04/2017
		07/04/2017	06/04/2018
	The calibration was conducted by accredited third parties which is Heyuan Power Supply Bureau was accredited by Guangdong Electric Power Group Corporation valid from 21/07/2009 to 20/07/2017 /13/.		
QA/QC procedure:	Data record will be archived for a period of 2 years after the crediting period to which the records pertain.		
Means of verification:	Data of the parameter was verified by checking MRRs and ETNs. All data is in line with MRRs and ETNs; Information flow was verified by checking MRRs and ETNs, and all information are consistent; Monitoring method was verified by site visit, checking calibration certificates, all monitoring method meets the description in the PDD; Calibration was verified by checking calibration certificate and Accreditation certificate, all calibration of monitoring equipment meets the requirement indicated in the PDD.		

Data / Parameter:	EG _{input,y}
Data unit:	MWh
Description:	Annual on-grid electricity purchased from CSPG by the proposed project for the plant operation
Purpose of the data:	Calculation of baseline emissions
Parameter value:	404.222
Source of data used:	Annual on-grid electricity purchased from CSPG by the proposed project for the plant operation (EG _{input,y}) are all sourced from Meter Reading Record

	(MRRs) /10/ issued by the project owner and Electricity Transaction Notes (ETNs) /11/ issued by power grid company covering monitoring period.																													
Information flow:	2 bidirectional electricity meters (M1 as main and M2 as backup) installed at the entrance of the Lankou 110kV Transformer Station and the exit of the booster station of the project respectively were measured continuously, recorded monthly and archived electronically. At 24:00 hr the last day of each month, the staff from project owner will record main electricity meter’s readings and form Meter Reading Records (MRRs). The staff from power grid company will record the meter readings of meters (Main meter) then transcribes the data into Electricity Transaction Notes (ETNs). The data for MRRs and ETNs have been sent to the CDM consulting company for reporting of GHG emission reduction. The conservative one would be used for ER calculation.																													
Monitoring method, frequency and equipments:	The parameter was measured continuously and recorded monthly by 2 bidirectional electricity meters (M1 as main and M2 as backup) installed at the entrance of the Lankou 110kV Transformer Station and the exit of the booster station of the project respectively during the monitoring period verified by site visit. See below for the information of 2 bidirectional electricity meters verified by site visit and checking calibration certificates /12/: <table><tr><th>Meter</th><th>Type</th><th>Serial Number</th><th>Accuracy</th></tr><tr><td>M1</td><td>DTSD341</td><td>01010801048</td><td>0.5s</td></tr><tr><td>M2</td><td>YD2103</td><td>070817234</td><td>0.5s</td></tr></table> The type, serial number and accuracy have been confirmed by site visit.	Meter	Type	Serial Number	Accuracy	M1	DTSD341	01010801048	0.5s	M2	YD2103	070817234	0.5s																	
Meter	Type	Serial Number	Accuracy																											
M1	DTSD341	01010801048	0.5s																											
M2	YD2103	070817234	0.5s																											
Calibration:	The calibration information is shown as below /12/: <table><tr><th>Meter</th><th>Calibration date</th><th>Valid until</th></tr><tr><td rowspan="6">M1</td><td>12/04/2012</td><td>11/04/2013</td></tr><tr><td>11/04/2013</td><td>10/04/2014</td></tr><tr><td>10/04/2014</td><td>09/04/2015</td></tr><tr><td>09/04/2015</td><td>08/04/2016</td></tr><tr><td>08/04/2016</td><td>07/04/2017</td></tr><tr><td>07/04/2017</td><td>06/04/2018</td></tr><tr><td rowspan="6">M2</td><td>12/04/2012</td><td>11/04/2013</td></tr><tr><td>11/04/2013</td><td>10/04/2014</td></tr><tr><td>10/04/2014</td><td>09/04/2015</td></tr><tr><td>09/04/2015</td><td>08/04/2016</td></tr><tr><td>08/04/2016</td><td>07/04/2017</td></tr><tr><td>07/04/2017</td><td>06/04/2018</td></tr></table>	Meter	Calibration date	Valid until	M1	12/04/2012	11/04/2013	11/04/2013	10/04/2014	10/04/2014	09/04/2015	09/04/2015	08/04/2016	08/04/2016	07/04/2017	07/04/2017	06/04/2018	M2	12/04/2012	11/04/2013	11/04/2013	10/04/2014	10/04/2014	09/04/2015	09/04/2015	08/04/2016	08/04/2016	07/04/2017	07/04/2017	06/04/2018
Meter	Calibration date	Valid until																												
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	09/04/2015	08/04/2016																												
	08/04/2016	07/04/2017																												
	07/04/2017	06/04/2018																												
M2	12/04/2012	11/04/2013																												
	11/04/2013	10/04/2014																												
	10/04/2014	09/04/2015																												
	09/04/2015	08/04/2016																												
	08/04/2016	07/04/2017																												
	07/04/2017	06/04/2018																												

	The calibration was conducted by accredited third parties which is Heyuan Power Supply Bureau was accredited by Guangdong Electric Power Group Corporation valid from 21/07/2009 to 20/07/2017 /13/.
QA/QC procedure:	Data record will be archived for a period of 2 years after the crediting period to which the records pertain.
Means of verification:	Data of the parameter was verified by checking MRRs and ETNs. All data is in line with MRRs and ETNs; Information flow was verified by checking MRRs and ETNs, and all information are consistent; Monitoring method was verified by site visit, checking calibration certificates, all monitoring method meets the description in the PDD; Calibration was verified by checking calibration certificate and Accreditation certificate, all calibration of monitoring equipment meets the requirement indicated in the PDD.

Data / Parameter:	TEG _y
Data unit:	MWh
Description:	Total electricity produced by the project activity, including the electricity supplied to the grid and the electricity supplied to internal loads, in year y
Purpose of the data:	Calculation of project emissions
Parameter value:	434,217.794
Source of data used:	Total electricity produced by the project activity, including the electricity supplied to the grid and the electricity supplied to internal loads, in year y (TEG _y) are all sourced from Meter Reading Record (MRRs) /10/ covering monitoring period.
Information flow:	3 electricity meters (M3, M4 and M5) installed at the exit of each generator of the project respectively were measured continuously, recorded monthly and archived electronically. At 24:00 hr the last day of each month, the staff from project owner will record electricity meters' readings and form Meter Reading Records (MRRs). The data for MRRs have been sent to the CDM consulting company for reporting of GHG emission reduction.
Monitoring method, frequency and equipments:	The parameter was measured continuously and recorded monthly by 3 electricity meters (M3, M4 and M5) installed at the exit of each generator of the project respectively during the monitoring period verified by site visit.

	See below for the information of 3 electricity meters verified by site visit and checking calibration certificates /12/: <table><tr><th>Meter</th><th>Type</th><th>Serial Number</th><th>Accuracy</th></tr><tr><td>M3</td><td>YD2103</td><td>070817236</td><td>0.5s</td></tr><tr><td>M4</td><td>YD2103</td><td>070817411</td><td>0.5s</td></tr><tr><td>M5</td><td>YD2103</td><td>070817233</td><td>0.5s</td></tr></table> The type, serial number and accuracy have been confirmed by site visit.	Meter	Type	Serial Number	Accuracy	M3	YD2103	070817236	0.5s	M4	YD2103	070817411	0.5s	M5	YD2103	070817233	0.5s																										
Meter	Type	Serial Number	Accuracy																																								
M3	YD2103	070817236	0.5s																																								
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Calibration:	The calibration information is shown as below /12/: <table><tr><th>Meter</th><th>Calibration date</th><th>Valid until</th></tr><tr><td rowspan="6">M3</td><td>12/04/2012</td><td>11/04/2013</td></tr><tr><td>11/04/2013</td><td>10/04/2014</td></tr><tr><td>10/04/2014</td><td>09/04/2015</td></tr><tr><td>09/04/2015</td><td>08/04/2016</td></tr><tr><td>08/04/2016</td><td>07/04/2017</td></tr><tr><td>07/04/2017</td><td>06/04/2018</td></tr><tr><td rowspan="6">M4</td><td>12/04/2012</td><td>11/04/2013</td></tr><tr><td>11/04/2013</td><td>10/04/2014</td></tr><tr><td>10/04/2014</td><td>09/04/2015</td></tr><tr><td>09/04/2015</td><td>08/04/2016</td></tr><tr><td>08/04/2016</td><td>07/04/2017</td></tr><tr><td>07/04/2017</td><td>06/04/2018</td></tr><tr><td rowspan="6">M5</td><td>12/04/2012</td><td>11/04/2013</td></tr><tr><td>11/04/2013</td><td>10/04/2014</td></tr><tr><td>10/04/2014</td><td>09/04/2015</td></tr><tr><td>09/04/2015</td><td>08/04/2016</td></tr><tr><td>08/04/2016</td><td>07/04/2017</td></tr><tr><td>07/04/2017</td><td>06/04/2018</td></tr></table> The calibration was conducted by accredited third parties which is Heyuan Power Supply Bureau was accredited by Guangdong Electric Power Group Corporation valid from 21/07/2009 to 20/07/2017 /13/.	Meter	Calibration date	Valid until	M3	12/04/2012	11/04/2013	11/04/2013	10/04/2014	10/04/2014	09/04/2015	09/04/2015	08/04/2016	08/04/2016	07/04/2017	07/04/2017	06/04/2018	M4	12/04/2012	11/04/2013	11/04/2013	10/04/2014	10/04/2014	09/04/2015	09/04/2015	08/04/2016	08/04/2016	07/04/2017	07/04/2017	06/04/2018	M5	12/04/2012	11/04/2013	11/04/2013	10/04/2014	10/04/2014	09/04/2015	09/04/2015	08/04/2016	08/04/2016	07/04/2017	07/04/2017	06/04/2018
Meter	Calibration date	Valid until																																									
M3	12/04/2012	11/04/2013																																									
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	08/04/2016	07/04/2017																																									
	07/04/2017	06/04/2018																																									
M4	12/04/2012	11/04/2013																																									
	11/04/2013	10/04/2014																																									
	10/04/2014	09/04/2015																																									
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M5	12/04/2012	11/04/2013																																									
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QA/QC procedure:	Data record will be archived for a period of 2 years after the crediting period to which the records pertain.																																										
Means of verification:	Data of the parameter was verified by checking MRRs. All data is in line with MRRs; Information flow was verified by checking MRRs, and all information are consistent; Monitoring method was verified by site visit, checking calibration certificates, all monitoring method meets the description in the PDD;																																										

	Calibration was verified by checking calibration certificate and Accreditation certificate, all calibration of monitoring equipment meets the requirement indicated in the PDD.
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Data / Parameter:	Cap _{PJ}
Data unit:	W
Description:	Installed capacity of the hydro power plant after the implementation of the project activity
Purpose of the data:	Calculation of project emissions
Parameter value:	260,000,000
Source of data used:	Installed capacity of the hydro power plant after the implementation of the project activity (Cap _{PJ}) is determined by the Nameplate of the equipment /9/ in this monitoring period.
Information flow:	Installed capacity of the hydro power plant after the implementation of the project activity (Cap _{PJ}) is determined by the Nameplate of the equipment /9/ in this monitoring period.
Monitoring method, frequency and equipments:	Installed capacity of the hydro power plant after the implementation of the project activity (Cap _{PJ}) is determined by the Nameplate of the equipment /9/ in this monitoring period.
Calibration:	Not applicable
QA/QC procedure:	Data record will be archived for a period of 2 years after the crediting period to which the records pertain.
Means of verification:	Data of the parameter was verified by checking Nameplate of the equipment. All data is in line with Nameplate of the equipment; Information flow was verified by checking Nameplate of the equipment, and all information are consistent.

Data / Parameter:	A _{PJ}
Data unit:	m ²
Description:	The area of the reservoir measured in the surface of the water, after the implementation of the project activity, when the reservoir is full
Purpose of the data:	Calculation of project emissions
Parameter value:	2012: 141,441

	2013: 141,438 2014: 141,440 2015: 141,441 2016: 141,443 2017: 141,441
Source of data used:	The area of the reservoir measured in the surface of the water, after the implementation of the project activity, when the reservoir is full (A_{PJ}) is determined by the Survey report issued by Guizhou Second Institute of Surveying and Mapping /14/ in this monitoring period.
Information flow:	The area of the reservoir measured in the surface of the water, after the implementation of the project activity, when the reservoir is full (A_{PJ}) is determined by the Survey report issued by Guizhou Second Institute of Surveying and Mapping /14/ in this monitoring period.
Monitoring method, frequency and equipments:	The area of the reservoir measured in the surface of the water, after the implementation of the project activity, when the reservoir is full (A_{PJ}) is determined by the Survey report issued by Guizhou Second Institute of Surveying and Mapping /14/ in this monitoring period.
Calibration:	Not applicable
QA/QC procedure:	Data record will be archived for a period of 2 years after the crediting period to which the records pertain.
Means of verification:	Data of the parameter was verified by checking Survey report issued by Guizhou Second Institute of Surveying and Mapping. All data is in line with Survey report issued by Guizhou Second Institute of Surveying and Mapping; Information flow was verified by checking Survey report issued by Guizhou Second Institute of Surveying and Mapping, and all information are consistent.

- Parameters available at validation stage:
- Below data has been verified against the data sources and the PDD.

Parameter title	Description	Data	Source
EF_{grid,CM,y^*}	Combined margin CO ₂ emission factor for CSPG in this monitoring period	0.8712	Registered PDD
Cap _{BL}	Install Capacity of the hydro power	0	ACM0002

	plant before the implementation of the project activity		
ABL	Area of the reservoir measured in the surface of the water, before the implementation of the project activity, when the reservoir is full	0	ACM0002

- * $EF_{grid,CM,y}$ is not directly available in the registered PDD but calculated based on the ex-ante data fixed in the registered PDD.

4.5 Quality of Evidence to Determine GHG Emission Reductions and Removals

- The monitoring has been carried out in accordance with CDM PDD /3/ dated 01/03/2009.
- As a result of verification of the ER calculation process, the assessment team confirmed that all the parameters required for the determination of the emission reductions have been included in the MR Report and ER Calculation Spreadsheet /2/ and are consistent with the applied methodology ACM0002 version 7.0 and the monitoring plan. The parameters are complete in this monitoring period.
- By checking the original record, crosscheck with the supporting evidence issued from other party than project owner, checking calibration related documents and interview with project owner through the site visit, the VVB is able to confirm there are no transposition errors between data sets. All data are consistent in all data sets.
- After verifying the reported figures with the raw data sources, it's confirmed that the values of the parameters from the raw data sources are consistent with those quoted in the ER Calculation Spreadsheet and the MR Report. The verification process for the same has been clearly described above in section 4.4 of the report.

4.6 Non-Permanence Risk Analysis

- Not applicable as a renewable project.

5 VERIFICATION CONCLUSION

- Applus+ Certification has been commissioned by Beijing 3050 Technology Co.,Ltd. to perform the verification of greenhouse gas emission reductions of the project activity “Guangdong Lankou 26MW Hydro Power Project” (VCS Ref. No. 371).
- The management of Shenzhen City Fuyuan Industry (Group) Co., Ltd. is responsible for the preparation of the GHG emissions data and the reported GHG emissions reductions on the basis set out within the project’s Monitoring Plan in the CDM PDD /3/ dated 01/03/2009.
- Our verification approach was based on the requirements as defined under the applicable VCS standards and relevant UNFCCC requirements. Our approach is risk-based, drawing on an understanding of the risks associated with reporting GHG emissions data and the controls in place to mitigate these. The verification can confirm that:
 - the project is implemented and operated as per the registered PDD;
 - the monitoring plan in the registered PDD is as per the applied methodology;
 - the monitoring complies with the registered PDD;
 - the monitoring report and other supporting documents provided are complete and verifiable and in accordance with the applicable VCS and CDM requirements;
 - the installed equipment being essential for generating emission reduction runs reliably and is calibrated appropriately;
 - the monitoring system is in place and generates GHG emission reductions data;
 - the GHG emission reductions are calculated without material misstatements.
- In our opinion, the GHG emission reductions for “Guangdong Lankou 26MW Hydro Power Project” during the monitoring period 01/06/2012 to 31/10/2017 as reported in Monitoring Report, prepared on the basis of the project’s Monitoring Plan are fairly stated. Based on the information we have seen and evaluated, we confirm the following statement:
 - Verification period: From 01/06/2012 to 31/10/2017 (divided into 6 vintage periods). Verified GHG emission reductions or removals in the above reporting period:

Year	Baseline emissions or removals (tCO ₂ e)	Project emissions or removals (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Net GHG emission reductions or removals (tCO ₂ e)
2012	31,237	0	0	31,237
2013	66,561	0	0	66,561
2014	58,376	0	0	58,376
2015	53,737	0	0	53,737
2016	90,156	0	0	90,156
2017	66,042	0	0	66,042
Total	366,109	0	0	366,109

APPENDIX 1: REFERENCE LIST

1. Monitoring report, version 01, dated 20/05/2022; version 02, dated 14/07/2022
2. ER calculation spreadsheet
3. Registered CDM PDD, version 5.0, dated 01/03/2009
4. Validation report, No. JCI-VAL 07/036, version 05, completed by JCI;
VCS Verification report, completed on 07/01/2010 by CEC
5. VCS standard version 4.3, dated on 22/06/2022
6. Statement issued by project owner
7. Approved methodology ACM0002, version 7.0, dated 14/12/2007
8. CDM Monitoring procedure
9. Nameplate of the equipment
10. Meter Reading Record (MRRs) for Meters
11. Electricity Transaction Notes covering the monitoring period
12. Calibration certificates of meters covering the whole monitoring period issued by Heyuan Power Supply Bureau
13. Accreditation certificates for Heyuan Power Supply Bureau issued by Guangdong Electric Power Group Corporation;
14. Survey report issued by Guizhou Second Institute of Surveying and Mapping

15 Staff roster of project

16 Payment Slip